

COMP 9200

Applied Research Methods

Methods and methodology

Overview

- Design of studies
- Variables and control
- Experimental Designs
 - Pre-Experimental Designs
 - True Experimental Designs
 - Quasi-Experimental Designs
 - Ex Post Facto Designs
 - Factorial Designs
- Data and measurement
 - Measurement
 - Validity
 - Reliability
 - Using statistics

Study design

- Correlational research
 - Examines extent to which differences in one variable/characteristic related to differences in one or more other ones
 - Researchers gather data about 2+ characteristics for particular group to determine any interrelationships
 - Correlational data plotted on scatter plot
 - Correlation $\neq$ causation!

Study design

- Cross-sectional study
 - People from different age groups sampled and compared
- Longitudinal study
 - Single group of people followed over time, and data related to characteristics under investigation collected at various times

Study design

- Cross-sectional:
 - easier to conduct because all data can be collected at one time
 - different age groups may represent different life experiences: threat to internal validity
 - correlations between characteristics at different age levels cannot be computed
- Longitudinal:
 - data collected over period of months or years
 - participants may drop out
 - people can get used to measurement instrument, and thus affect their response

Study design

- Cohort-Sequential:
 - Addresses some weaknesses of longitudinal and cross-sectional designs
 - Researcher begins with 2+ groups (like cross-sectional) and follows each group over period of time (like longitudinal piece)
 - Allows calculation of correlations between measures taken at 2 different times: predictions can be made across time

Independent / Dependent Variables

- **Variable:** quality/characteristic in research investigation with 2+ values
- **Cause-and-Effect Relationship:** how much one variable (cause) influences another variable (effect)
- **Independent Variable:** variable that researcher studies as possible cause of something else (researcher directly manipulates)
- **Dependent Variable:** variable potentially influenced by independent variable (influenced by / depends on independent variable)

Control

- **Internal Validity:**

- How much does research study and data it yields allows researcher to draw accurate conclusions about cause-and-effect and other relationships
- Without internal validity in experimental designs, results are not interpretable

- **Confounding Variables:**

- account for differences in 2+ groups that are not attributable to what is being studied

Confounding variables

- Keep some things constant
- Include a control group
- Randomly assign people to groups
- Assess equivalence before the treatment with one or more pretests
- Expose participants to all experimental conditions
- Statistically control for confounding variables

Experimental Designs

- Pre-Experimental Designs
- True Experimental Designs
- Quasi-Experimental Designs
- Ex Post Facto Designs
- Factorial Designs

Pre-Experimental Designs

- Not possible to show cause-and-effect relationships because
 - independent variable does not vary or
 - experimental and control groups are not comprised of equivalent or randomly selected individuals

Pre-Experimental Designs

Design 1: One-Shot Experimental Case Study (low internal validity)

Group 1	Tx	Obs
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Design 2: One-Group Pretest-Posttest Design

Group 1	Obs	Tx	Obs
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Design 3: Static Group Comparison

Group 1	Tx	Obs
Group 2	---	Obs

True Experimental Designs

Compared to pre-experimental designs, offer great degree of control and internal validity

Design 4: Pretest-Posttest Control Group Design (random assignment)

Design 5: Solomon Four-Group Design (enhances external validity)

Design 6: Within-Subjects Design (repeated measures)

Quasi-Experimental Designs

Randomness is not possible or practical; can't control for all confounding variables

Design 8: Nonrandomized Control Group Pretest-Posttest Design

Design 9: Simple Time-Series Design

Design 10: Control Group, Time-Series Design

Quasi-Experimental Designs

Design 11: Reversal Time-Series Design

Design 12: Alternating Treatment Design

Design 13: Multiple Baseline Design

Ex Post Facto Designs

Researcher identifies events that have already occurred or conditions that are already present and then collects data to investigate a possible relationship between these factors and subsequent characteristics or behaviors

- like *correlational research*, involves looking at existing circumstances
- like *experimental research*, has clearly identifiable independent and dependent variables
- unlike *experimental research*, involves no direct manipulation of independent variable – the presumed “cause” has already occurred

Design 14: Simple Ex Post Facto Design

Factorial Designs

Examination of effects of two or more independent variables in a single study

Design 15: Randomized Two-Factor Design

Design 16: Combined Experimental and Ex Post Facto Design

Measurement

- Limiting data of any phenomenon
 - Interpretation
 - Comparison with standard
- Substantial measurements
 - Have physical substance
- Insubstantial measurements
 - Concepts and ideas
 - Opinions
 - Other intangible entities

Measurement

- Four scales of measurement
- Nominal
- Ordinal
- Interval
- Ratio

Measurement

- Nominal
- Assign names to data
- Infinite number of ways
- Simplistic
- Divides data into discrete categories
- Statistical procedures = mode, percentage, chi-square test

Measurement

- Ordinal
- Think in terms of symbols ($>$; $<$)
- Allows data to be rank-ordered
- Statistical procedures = median, percentile rank, Spearman's rank-order correlation

Measurement

- Interval
- Equal units of measurement
- Zero point established arbitrarily
- Rating scales, such as surveys, assumed to fall on interval scales
- Statistical procedures = means, standard deviations, Pearson product moment correlations

Measurement

- Ratio
- Equal measurement units
- Has absolute zero point
- Can express values in terms of multiples and fractional parts
- Relatively rare outside the physical sciences

Measurement

- Summary
- Nominal scale: A is different from B
- Ordinal scale: A is bigger/better/more of something than B
- Interval scale: A is so many units (degrees/inches) more than B
- Ratio scale: A is so many times as big/bright/tall/heavy as B

Validity and Reliability

- Validity
- Extent to which a measurement instrument measures what it is intended to measure
- Reliability
- Consistency with which a measurement instrument yields a certain result when the entity being measured has not changed.

Review

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